# Winter Consulting IIT Guwahati - Dec 2025 - Jan 2026

## CASE 2 - Viewer Retention in OTT Platforms: Diagnosing Engagement Patterns

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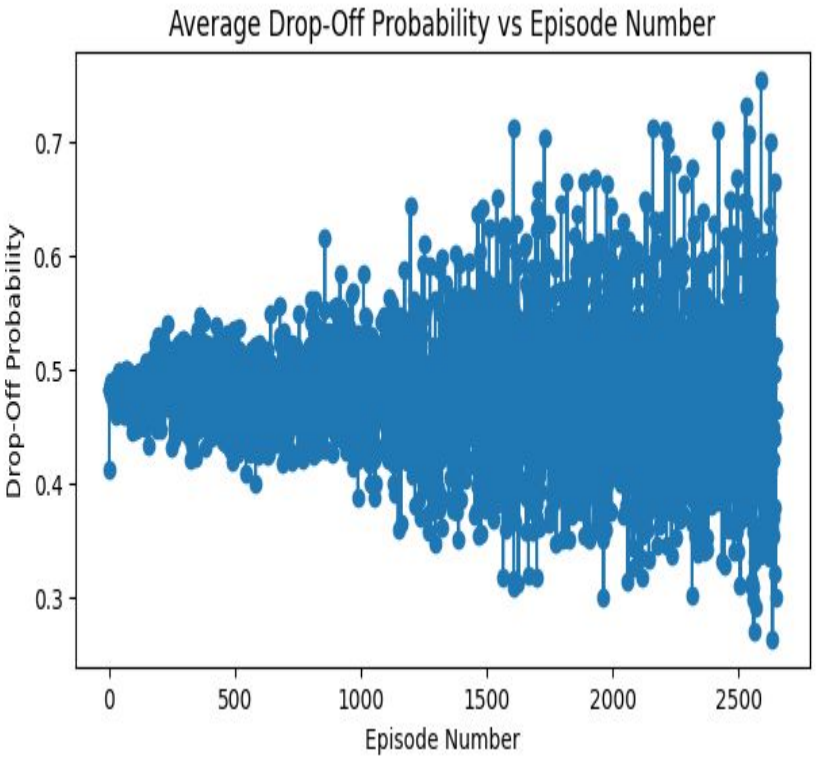
# Business impact and analytical objective

## Early Viewer Drop-off in Season 1 based on episode number

Significant viewer disengagement during Season 1 reduces lifetime value and complicates renewal decisions.

Hypothesis: episode-level design choices (pacing, cognitive load, structure) drive early drop-off.

Objective: identify drivers of drop-off, segment episodes into actionable groups, and recommend feasible content and product interventions.



# Drivers of Viewer Continuation

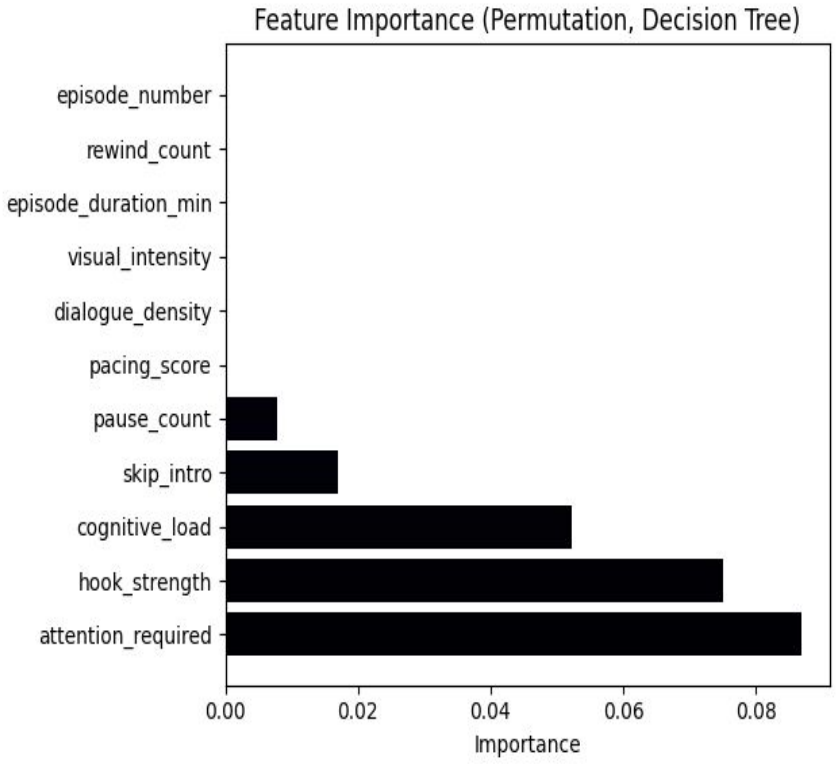
## Model findings and predictive strength

**Top predictors:** pacing_score, hook_strength, attention_required.

Episodes with strong hooks in the first 3–5 minutes and balanced pacing show materially higher continuation.

Dense dialogue combined with high attention_required correlates with elevated drop-off.

Models show outstanding discrimination (ROC-AUC ≈ 0.98–0.99), supporting reliable identification of high-risk episodes.



# Episode Segmentation via Clustering

## Distinct episode and engagement outcomes

K Means (k=4) on content and behavioral features produced four actionable clusters.

Cluster 0 — Benchmark: high pacing (6.01), strong hooks (7.02) → highest avg watch % (70.65), low drop-off (0.00).

Cluster 1 — Risk: low pacing (3.83), high cognitive load (7.68), higher attention_required (3) → highest drop-off (0.43), lowest avg watch % (44.65).

Cluster 2 — Mixed: moderate pacing (5.93), moderate hooks (5.52), highest dialogue_density (3.00) → mixed outcomes (avg watch 54.43, drop_off 0.12).

Cluster 3 — Niche: high pacing (6.03), weaker hooks (4.00), high visual_intensity → moderate avg watch % (58.65), low drop-off (0.00).

# Conclusions

## What the data proves and the business implication

**Content design choices** — pacing, hook strength, cognitive load — are **strong, consistent predictors** of early drop-off (models ROC-AUC ≈ 0.98–0.99).

**Clustering validates** that episodes fall into **distinct archetypes** with different engagement outcomes (e.g., Cluster 0 avg watch **70.65%** vs Cluster 1 avg watch **44.65%**, drop-off **0.43)**..

**Targeted interventions** on high-risk clusters (primarily Cluster 1) can **improve retention with limited creative disruption** and high ROI potential.

| Feature | Impact | Business implication |
|---|---|---|
| pacing_score | Higher pacing -> lower drop-off | Optimize pacing targets |
| hook_strength | Stronger hook -> higher continua | Prioritize hook edits |
| attention_required | Higher attention -> higher drop-c | Reduce cognitive load or add recap |
| cognitive_load | Higher cognitive load -> higher d | Introduce recaps, chapter markers |
| dialogue_density | Higher dialogue -> higher cogniti | Use on-screen cues, recaps, or pac |

# Prioritization Roadmap

## Sequencing actions by impact and feasibility

Priority 1 High impact, high feasibility: Hook optimization pilot; Skip Intro nudges. Rationale: low production cost, fast A/B testing.

Priority 2 High impact, medium feasibility: Cognitive load mitigations and recommendation tuning. Rationale: requires UX/product and editorial work.

Priority 3 Medium impact, lower feasibility: Creative playbooks and production guidelines. Rationale: longer lead time but sustainable.

Execution: run 2–3 concurrent pilots with clear success criteria and decision gates.

# Risks and Mitigation

## Monitoring and contingency plans for pilots

**Risk:** Creative changes may harm storytelling or alienate core fans. **Mitigation:** Pilot on a small sample; A/B test; collect qualitative feedback.

**Risk:** Product nudges may annoy users or reduce content exposure. **Mitigation:** Personalize nudges to high-risk segments; monitor opt-out and engagement metrics.

**Risk:** Resource constraints for editorial redesign. **Mitigation:** Prioritize Cluster 1 episodes with highest ROI; use automated tooling for chapter markers and recaps.

**Monitoring framework:** weekly KPIs for pilots (avg_watch_percentage, drop_off at 5/10/20 minutes, retention cohort lift); decision gate at 4–6 weeks.

| Risk | Likelihood | Impact | Mitigation | Owner |
|---|---|---|---|---|
| Creative disruption | Medium | High | Pilot + A/B test | Content Ops |
| Nudge annoyance | Medium | Medium | Personalize nudges | Product |
| Resource constraints | High | Medium | Prioritize high ROI episodes | Editorial |

# Thank you

**Immediate next step:** Run 4–6 week pilots on Cluster 1 (Hook optimization + Skip-Intro nudges); review KPI gates at week 4–6.

**Deliverables available:** Full analysis notebook, model artifacts.

**Decision request:** Approve pilot scope and sample episodes so we can finalize A/B test design and success criteria.